

17.3 Constrained models

The cell-means model assumes nothing about the table of means. Smaller models, such as the additive model or a three-way model without the three-factor interaction, are the cell-means model with linear constraints $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$. This section fits and tests them with the restricted least squares of [Section 11.3](#), and shows that main-effect hypotheses can merge once the constraints hold.

17.3.1 Estimation and tests under constraints

Theorem 17.3.1 — Constrained cell-means models

In the cell-means model of Definition 17.1.1, let $\mathbf{R}$ be a $g \times p$ matrix of rank g , and let the columns of the $p \times (p - g)$ matrix $\mathbf{K}$ form a basis of $\mathcal{N}(\mathbf{R})$. Assume that $\mathbf{WK}$ has full column rank $p - g$. The **constrained model** is $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\mu}$ with $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$.

- (a) The constrained model is the full-rank linear model $E(\mathbf{Y}) = \mathbf{WK}\boldsymbol{\theta}$, with $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$. Every cell mean, including the means of empty cells, is estimable in it. The least squares estimate is

$$\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \mathbf{V}_{\mathbf{R}}\mathbf{D}\bar{\mathbf{y}}, \quad \mathbf{V}_{\mathbf{R}} = \mathbf{K}(\mathbf{K}^{\top}\mathbf{D}\mathbf{K})^{-1}\mathbf{K}^{\top}, \quad \text{Cov}(\hat{\boldsymbol{\mu}}_{\mathbf{R}}) = \sigma^2\mathbf{V}_{\mathbf{R}}, \quad (17.3.1)$$

and its residual sum of squares $\text{SSE}_{\mathbf{R}}$ has $n - p + g$ degrees of freedom.

- (b) If every cell is occupied, then $\mathbf{V}_{\mathbf{R}} = \mathbf{D}^{-1} - \mathbf{D}^{-1}\mathbf{R}^{\top}(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^{\top})^{-1}\mathbf{R}\mathbf{D}^{-1}$, so $\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \bar{\mathbf{y}} - \mathbf{D}^{-1}\mathbf{R}^{\top}(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^{\top})^{-1}\mathbf{R}\bar{\mathbf{y}}$, and $\text{SSE}_{\mathbf{R}} - \text{SSE} = (\mathbf{R}\bar{\mathbf{y}})^{\top}(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^{\top})^{-1}\mathbf{R}\bar{\mathbf{y}}$ is the sum of squares, on g degrees of freedom, of the hypothesis $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ in the unconstrained model.
- (c) Under normality, for an $\ell \times p$ matrix $\mathbf{L}$ with $q = \text{rank}(\mathbf{L}\mathbf{K}) \geq 1$, the F statistic for $H : \mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ within the constrained model is

$$F = \frac{(\mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}})^{\top}(\mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^{\top})^{-}\mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}}}{q s_{\mathbf{R}}^2}, \quad s_{\mathbf{R}}^2 = \frac{\text{SSE}_{\mathbf{R}}}{n - p + g}, \quad (17.3.2)$$

for any generalized inverse. It has the $F(q, n - p + g, \gamma)$ distribution with $\gamma = (\mathbf{L}\boldsymbol{\mu})^{\top}(\mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^{\top})^{-}\mathbf{L}\boldsymbol{\mu}/\sigma^2$.

- (d) Within the constrained model, $\mathbf{L}_1\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{L}_2\boldsymbol{\mu} = \mathbf{0}$ are the same hypothesis iff $\mathbf{L}_1\mathbf{K}$ and $\mathbf{L}_2\mathbf{K}$ have the same row space, and then their F statistics coincide.

Proof.

- (a) A table satisfies $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ iff $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$ for a unique $\boldsymbol{\theta}$, so the constrained model is the linear model with model matrix $\mathbf{WK}$, of full column rank by assumption. Every coordinate of $\boldsymbol{\theta}$ is then estimable, and so is every linear function of $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$. Since $(\mathbf{WK})^{\top}\mathbf{WK} = \mathbf{K}^{\top}\mathbf{D}\mathbf{K}$ and $\mathbf{W}^{\top}\mathbf{y} = \mathbf{D}\bar{\mathbf{y}}$, the least squares estimate is $\hat{\boldsymbol{\theta}} = (\mathbf{K}^{\top}\mathbf{D}\mathbf{K})^{-1}\mathbf{K}^{\top}\mathbf{D}\bar{\mathbf{y}}$, with covariance $\sigma^2(\mathbf{K}^{\top}\mathbf{D}\mathbf{K})^{-1}$ (Theorem 5.5.1). Multiplying by $\mathbf{K}$ gives [Eq. 17.3.1](#). The model matrix has rank $p - g$, which gives the degrees of freedom.
- (b) With every cell occupied, $\mathbf{W}$ has full column rank, $\mathbf{D}^{-1}$ is the inverse of $\mathbf{W}^{\top}\mathbf{W}$,

and $\bar{\mathbf{y}}$ is the unrestricted estimate. The constrained model is the reduced model of the testable hypothesis $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ in the unconstrained one. So Proposition 11.3.3(a) and (b), with $\mathbf{A}^\top = \mathbf{R}$ and $\mathbf{G} = \mathbf{D}^{-1}$, give the displayed formula for $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$ and $\text{SSE}_{\mathbf{R}} = \text{SSE} + (\mathbf{R}\bar{\mathbf{y}})^\top (\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1} \mathbf{R}\bar{\mathbf{y}}$, the sum of squares Eq. 17.1.1 of the hypothesis. For the covariance, let $\mathbf{V}'$ be the right side of the claimed identity. On vectors $\mathbf{D}\mathbf{K}\mathbf{a}$ both matrices give $\mathbf{K}\mathbf{a}$, because $\mathbf{R}\mathbf{K} = \mathbf{0}$. On vectors $\mathbf{R}^\top \mathbf{b}$ both give $\mathbf{0}$, because $\mathbf{K}^\top \mathbf{R}^\top = \mathbf{0}$ and $\mathbf{V}'\mathbf{R}^\top \mathbf{b} = \mathbf{D}^{-1}\mathbf{R}^\top \mathbf{b} - \mathbf{D}^{-1}\mathbf{R}^\top \mathbf{b}$. These vectors span $\mathbb{R}^p$: the spaces $\mathcal{C}(\mathbf{D}\mathbf{K})$ and $\mathcal{C}(\mathbf{R}^\top)$ have dimensions $p - g$ and g , and if $\mathbf{D}\mathbf{K}\mathbf{a} = \mathbf{R}^\top \mathbf{b}$, then $\mathbf{a}^\top \mathbf{K}^\top \mathbf{D}\mathbf{K}\mathbf{a} = (\mathbf{R}\mathbf{K}\mathbf{a})^\top \mathbf{b} = 0$, so $\mathbf{K}\mathbf{a} = \mathbf{0}$ because $\mathbf{D}$ is positive definite. Two matrices that agree on a spanning set are equal.

- (c) In the full-rank model $\mathbf{W}\mathbf{K}\boldsymbol{\theta}$ the hypothesis is $\mathbf{L}\mathbf{K}\boldsymbol{\theta} = \mathbf{0}$, which is testable. Theorem 11.3.2 with $\mathbf{A}^\top = \mathbf{L}\mathbf{K}$ and $\mathbf{G} = (\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1}$ gives the statistic, since $\mathbf{L}\mathbf{K}\boldsymbol{\theta} = \mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}}$ and $\mathbf{L}\mathbf{K}(\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{L}^\top = \mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^\top$.
- (d) Within the model, $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ iff $\boldsymbol{\theta} \in \mathcal{N}(\mathbf{L}\mathbf{K})$, and two matrices have the same null space iff they have the same row space. The hypotheses then have the same reduced model, and by Theorem 11.3.2(b) the same F statistic.

■

Part (d) is the new phenomenon: hypotheses that differ in the cell-means model can coincide once the constraints hold. Part (b) pulls the cell averages onto the constraint set along the directions $\mathbf{D}^{-1}\mathbf{R}^\top$. The rank assumption on $\mathbf{W}\mathbf{K}$ holds whenever every cell is occupied; with empty cells it is a real condition (Exercise (C2)).

17.3.2 The additive model

Take $\mathbf{R} = \mathbf{C}_a^\top \otimes \mathbf{C}_b^\top$, the interaction hypothesis matrix of Eq. 17.1.2, with $g = (a-1)(b-1)$. Its null space is the space of additive tables $\mu_{ij} = \alpha_i + \beta_j$ (Section 17.1), so the constrained model is the additive model Eq. 8.6.1 written in cell means.

Corollary 17.3.2 — Main effects in the additive model

In the two-way layout with all cells occupied, let $\mathbf{R} = \mathbf{C}_a^\top \otimes \mathbf{C}_b^\top$. Then:

- (a) $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$ is the table of fitted cell means of the additive model, and $\text{SSE}_{\mathbf{R}} - \text{SSE} = \text{SS}(AB \mid \mu, A, B)$;
- (b) for any weights $\mathbf{w} \in \mathbb{R}^b$ with $\sum_j w_j \neq 0$, the hypothesis that $\sum_j w_j \mu_{ij}$ is the same for all i is, within the additive model, the hypothesis $\alpha_1 = \cdots = \alpha_a$. Its sum of squares is $\text{SS}(A \mid \mu, B)$, whatever $\mathbf{w}$;
- (c) weights that differ from row to row do not have this property: the hypothesis that $\sum_j w_{ij} \mu_{ij}$ is the same for all i depends on the column effects unless $\sum_j w_{ij} \beta_j$ does not depend on i .

Proof.

- (a) The constrained mean vectors $\mathbf{W}\boldsymbol{\mu}$ with $\boldsymbol{\mu}$ additive form $\mathcal{C}([\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B])$, so the constrained fit is the additive fit. The difference of residual sums of squares is then $\text{SS}(AB \mid \mu, A, B)$ by definition. (b) For an additive table, $\sum_j w_j \mu_{ij} = (\sum_j w_j) \alpha_i +$

$\sum_j w_j \beta_j$, which is the same for all i iff the α_i are equal. So all these hypotheses define the same subspace of the constrained model, and by [Theorem 17.3.1\(d\)](#) the same test. Taking $\mathbf{w} = \mathbf{e}_1$ shows that the subspace is $\{\mu_{ij} = \beta_j\}$, the model with B only, and the drop in residual sum of squares from it to the additive model is $SS(A \mid \mu, B)$. (c) Now $\sum_j w_{ij} \mu_{ij} = (\sum_j w_{ij}) \alpha_i + \sum_j w_{ij} \beta_j$, and the second term varies with i in general. ■

So once additivity is accepted, the argument about types disappears for the main effects: any common weighting, and even a single column, gives the comparison $\alpha_i - \alpha_k$, tested by the Type II sum of squares (and by Type III, which coincides with it there). Only the count-weighted means of Type I stay contaminated, by part (c). The usual summaries after an additive fit are the **least squares means** (estimated marginal means), the unweighted row means of $\hat{\mu}_{\mathbf{R}}$, with standard errors from $\mathbf{V}_{\mathbf{R}}$.

Example. (The variety trial under additivity)

In Example (Five tables for one data set) the interaction sum of squares 2.146 gave $F = 1.71$ on 4 and 28 degrees of freedom, $p = 0.176$. Suppose we accept additivity. The constrained fit [Eq. 17.3.1](#) gives the table of fitted means

	S1	S2	S3
V1	5.56	7.17	7.95
V2	5.11	6.73	7.51
V3	4.48	6.09	6.87

with $SSE_{\mathbf{R}} = 10.927$ on $37 - 9 + 4 = 32$ degrees of freedom and $s_{\mathbf{R}}^2 = 0.3415$. The fitted mean of V1 at S3 is 7.95, although its two plots averaged 7.45: additivity carries V1's advantage at S1 and S2 over to S3. The test of equal unweighted variety means, [Eq. 17.3.2](#), gives 6.184, the Type II sum of squares, with $F = 9.06$ on 2 and 32 degrees of freedom, $p = 0.00076$. The script checks that site weights 0.5, 0.3, 0.2, and the column S1 alone, give the same 6.184 ([Corollary 17.3.2\(b\)](#)), while each variety's own count weights give 0.571, the Type I value. The least squares means are 6.894, 6.449 and 5.811 (standard errors 0.169, 0.170, 0.178), and V1 exceeds V2 by 0.445 (standard error 0.243).

The generating means had a small interaction: V1's advantage over V2 is 0.8 at S1, 0.7 at S2 and 0.1 at S3, and the test on 4 and 28 degrees of freedom has little power to detect it. A non-significant interaction test is not evidence of additivity.

```
import numpy as np
import pandas as pd
from scipy import stats
from scipy.linalg import null_space
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
```

```

rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
         for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])

cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().to_numpy()
n_c = cells.size().to_numpy()
Dinv = np.diag(1 / n_c)
C3 = np.array([[1.0, 0.0], [0.0, 1.0], [-1.0, -1.0]])
G = np.kron(C3.T, C3.T)
mu = 0
K = null_space(G)

V_G = K @ np.linalg.inv(K.T @ np.diag(n_c) @ K) @ K.T
sigma^2
mu_G = V_G @ np.diag(n_c) @ ybar
estimate
mu_G_direct = ybar - Dinv @ G.T @ np.linalg.solve(G @ Dinv @ G.T, G @ ybar)
print("additive fitted cell means:\n", mu_G.reshape(3, 3).round(3))

n, m, g = n_c.sum(), 9, G.shape[0]
sse = ((trial["y"] - cells.transform("mean")) ** 2).sum()
Gy = G @ ybar
ss_constraint = Gy @ np.linalg.solve(G @ Dinv @ G.T, Gy)
sse_G = sse + ss_constraint
s2_G = sse_G / (n - m + g)

def constrained_test(L):
    """F test of L mu = 0 within the constrained model G mu = 0."""
    u = L @ mu_G
    V = L @ V_G @ L.T
    ss = u @ np.linalg.pinv(V) @ u
    q = np.linalg.matrix_rank(L @ K)
    F = ss / q / s2_G
    return ss, q, F, stats.f.sf(F, q, n - m + g)

avg3 = np.ones((1, 3)) / 3
print("variety, unweighted:", np.round(constrained_test(np.kron(C3.T, avg3)), 4))
print("site, unweighted:", np.round(constrained_test(np.kron(avg3, C3.T)), 4))

```

[Run this code online](#)

```

lsmeans = (mu_G.reshape(3, 3)).mean(axis=1)
L_rows = np.kron(np.eye(3), avg3)
se_ls = np.sqrt(np.diag(L_rows @ V_G @ L_rows.T) * s2_G)
d = np.array([1.0, -1.0, 0.0]) @ L_rows
diff12, se_diff12 = d @ mu_G, np.sqrt(d @ V_G @ d * s2_G)
print("least squares means:", lsmeans.round(3), " se", se_ls.round(3))
print(f"V1 - V2: {diff12:.3f} (se {se_diff12:.3f})")

```

[Run this code online](#)

In the listing the constraint matrix is called `G`. The script checks the fit, the identity of Theo-

rem 17.3.1(b) and the tests against statsmodels.

17.3.3 Several factors: choosing among constrained models

With three or more factors the useful constrained models are the hierarchical ones, named by their highest interactions. In an $a \times b \times d$ layout, where C has d levels, $[AB][C]$ contains the AB interaction, the main effect of C , and everything below them. The model $[AB][AC][BC]$ sets the $(a-1)(b-1)(d-1)$ contrasts $\mathbf{C}_a^\top \otimes \mathbf{C}_b^\top \otimes \mathbf{C}_d^\top$ to zero. Unbalanced data make these models non-orthogonal, so we remove one term at a time, judging each step by its drop in residual sum of squares over the error mean square of the full model.

Example. (Simplifying a three-factor curing study)

A (synthetic) study of a resin coating measured hardness after curing at three temperatures (T), with two hardeners (H) and resin from three suppliers (S). The 18 cells received one to three panels each, 37 in all, so the full model $[THS]$ leaves 19 error degrees of freedom, with mean square 0.408. The backward path:

Step	From	To	Term removed	df	SS	F	p
1	$[THS]$	$[TH][TS][HS]$	THS	4	1.137	0.70	0.6036
2	$[TH][TS][HS]$	$[TH][HS]$	TS	4	1.845	1.13	0.3719
3	$[TH][HS]$	$[TH][S]$	HS	2	4.132	5.06	0.0173
4	$[TH][HS]$	$[T][HS]$	TH	2	13.716	16.81	0.0001
5	$[TH][HS]$	$[HS]$	T and TH	4	19.100	11.70	0.0001

Steps 1 and 2 cost little. Step 1 is the test of the constraints $\mathbf{G}_3\boldsymbol{\mu} = \mathbf{0}$ with $\mathbf{G}_3 = \mathbf{C}_3^\top \otimes \mathbf{C}_2^\top \otimes \mathbf{C}_3^\top$, and the script recovers its sum of squares from the cell means through Theorem 17.3.1(b). Steps 3 and 4 show that neither interaction with the hardener can go, so the path stops at $[TH][HS]$. In that model the effect of temperature under a given hardener is the same for all three suppliers, so whether temperature matters at all is a single question inside the model, answered by step 5.

```
import statsmodels.formula.api as smf
rng3 = np.random.default_rng(31)
levels = [(i, j, k) for i in range(3) for j in range(2) for k in range(3)]
n3 = rng3.integers(1, 4, size=len(levels)) # 1 to 3 runs per cell

def true_hardness(i, j, k):
    """Temperature i, hardener j, supplier k: hardener interacts with both, nothing else does."""
    return (40 + [0, 1.0, 2.0][i] + 1.5 * j + [0, 0.5, -0.5][k]
            + j * [0, -1.0, -2.0][i] + j * [0, 1.2, -0.8][k])

cure = pd.DataFrame([(f"t{i + 1}", f"h{j + 1}", f"s{k + 1}", true_hardness(i, j, k)
                      + rng3.normal(0, 0.7))
                     for (i, j, k), r in zip(levels, n3) for _ in range(r)],
```

```

columns=["temp", "hardener", "supplier", "y"])

def fit(formula):
    f = smf.ols(formula, cure).fit()
    return f.ssr, int(f.df_resid)

sse_full, df_full = fit("y ~ temp * hardener * supplier")
mse_full = sse_full / df_full
path = [("THS)", "[TH] [TS] [HS]", "y ~ temp * hardener * supplier", "y ~ (temp +
    hardener + supplier) ** 2"),
        ("[TH] [TS] [HS]", "[TH] [HS]", "y ~ (temp + hardener + supplier) ** 2", "y ~
    temp * hardener + hardener * supplier"),
        ("[TH] [HS]", "[TH] [S]", "y ~ temp * hardener + hardener * supplier", "y ~
    temp * hardener + supplier"),
        ("[TH] [HS]", "[T] [HS]", "y ~ temp * hardener + hardener * supplier", "y ~
    temp + hardener * supplier"),
        ("[TH] [HS]", "[HS]", "y ~ temp * hardener + hardener * supplier", "y ~
    hardener * supplier")]
steps = []
for big, small, f_big, f_small in path:
    (s1, d1), (s0, d0) = fit(f_big), fit(f_small)
    F = (s0 - s1) / (d0 - d1) / mse_full
    steps.append((big, small, d0 - d1, s0 - s1, F, stats.f.sf(F, d0 - d1, df_full)))
print(f"[big:13s] -> [small:13s] df = {d0 - d1}  SS = {s0 - s1:6.3f}  F = {F:5.2
f}  p = {steps[-1][5]:.4f}")

```

[Run this code online](#)

Such comparisons can look contradictory: a factor may matter alone but not after another. That is the unbalanced analogue of collinearity, and it should be reported as such, not resolved by choosing a type of sum of squares.

17.3.4 Exercises

17.3.4.1 A. Check your understanding

Exercise A1.

In an $a \times b \times d$ layout with every cell occupied and n observations, how many constraints define the model $[AB][AC][BC]$, and how many error degrees of freedom does it have? Check your answer against step 1 of [Example \(Simplifying a three-factor curing study\)](#).

17.3.4.2 B. Practice

Exercise B1.

For a 2×2 layout with all cells occupied, take $\mathbf{R} = (1, -1, -1, 1)$. Use [Theorem 17.3.1\(b\)](#) to show that the additive fit is $\hat{\mu}_{ij} = \bar{y}_{ij} - (-1)^{i+j} \hat{\delta} / (n_{ij}H)$, where $\hat{\delta} = \bar{y}_{11} - \bar{y}_{12} - \bar{y}_{21} + \bar{y}_{22}$ and $H = \sum_{ij} 1/n_{ij}$. Which cells move most?

Solution. Here $\mathbf{R}\bar{\mathbf{y}} = \hat{\delta}$ and $\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top = H$, so $\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \bar{\mathbf{y}} - \mathbf{D}^{-1}\mathbf{R}^\top \hat{\delta} / H$. The (i, j) entry of $\mathbf{D}^{-1}\mathbf{R}^\top$ is $(-1)^{i+j} / n_{ij}$. Each cell average moves by $\hat{\delta} / (n_{ij}H)$ in the direction that

removes the interaction, so the least replicated cells move most. The four moves remove $\hat{\delta}$ exactly: $\sum_{ij} (-1)^{i+j} \cdot (-1)^{i+j} \hat{\delta} / (n_{ij} H) = \hat{\delta}$.

Exercise B2.

In a 2×3 layout with every cell occupied, impose only $\mu_{11} - \mu_{12} - \mu_{21} + \mu_{22} = 0$. Describe the constrained model in words. How many degrees of freedom does its residual have, and what does the F test of the remaining interaction contrast within it test?

Exercise B3.

In the additive model with all cells occupied, show that the least squares means satisfy $\hat{\mu}_{i\cdot} - \hat{\mu}_{\cdot k} = \hat{\alpha}_i - \hat{\alpha}_k$ for any solution of the normal equations, and that the same difference is obtained if the fitted table is averaged over j with any fixed weights summing to one. Why does the choice of weights still matter for the least squares means themselves?

Solution. The fitted table is $\hat{\mu}_{ij} = \hat{m} + \hat{\alpha}_i + \hat{\beta}_j$, unique as a table (Theorem 6.4.2). Averaging over j with weights w_j summing to one gives $\hat{m} + \hat{\alpha}_i + \sum_j w_j \hat{\beta}_j$, and the last term does not depend on i . So the differences are $\hat{\alpha}_i - \hat{\alpha}_k$, an estimable function whose value does not depend on the solution chosen. The means themselves contain $\sum_j w_j \hat{\beta}_j$, which does depend on the weights. They are levels, not comparisons, and a level is a mean over some population of columns.

17.3.4.3 C. Going deeper

Exercise C1.

The *method of unweighted means* estimates $\alpha_1 - \alpha_2$ in a $2 \times b$ additive layout by $b^{-1} \sum_j (\bar{y}_{1j} - \bar{y}_{2j})$, as if every cell had the same count. Show that the least squares estimate is $\sum_j h_j (\bar{y}_{1j} - \bar{y}_{2j}) / \sum_j h_j$ with $h_j = n_{1j} n_{2j} / n_{\cdot j}$. Deduce that both are unbiased under additivity, and that the least squares estimate has the smaller variance unless all h_j are equal.

Solution. Under additivity each $\bar{y}_{1j} - \bar{y}_{2j}$ is unbiased for $\delta = \alpha_1 - \alpha_2$, with variance σ^2/h_j (Exercise (B2)), and the b differences are independent. Any weighted average with weights summing to one is unbiased. Its variance $\sigma^2 \sum_j v_j^2/h_j$ is minimized, by the Cauchy–Schwarz argument of Proposition 17.1.2, at $v_j \propto h_j$. The minimum variance is $\sigma^2/\sum_j h_j$. The least squares estimate is the best linear unbiased estimator (Theorem 7.2.1), and it is a linear function of the cell averages, which are sufficient here. So it is the h -weighted average. The unweighted average has $v_j = 1/b$, and it is optimal only if all h_j are equal.

Exercise C2.

Allow empty cells. Show that **WK** in Theorem 17.3.1 has full column rank iff no nonzero table satisfying $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ vanishes on every occupied cell. Deduce that for the additive constraints this holds iff the design graph of Theorem 8.6.1 is connected.

Solution. $\mathbf{WK}\boldsymbol{\theta} = \mathbf{0}$ iff the table $\mathbf{K}\boldsymbol{\theta}$ vanishes on every occupied cell, since the rows of $\mathbf{W}$ pick out the occupied cells. The columns of $\mathbf{K}$ are independent, so $\mathbf{WK}$ has a nonzero null vector iff some nonzero table satisfying the constraints vanishes on O . For additive tables $\alpha_i + \beta_j$, vanishing on O means $\alpha_i = -\beta_j$ for every occupied (i, j) . As in the proof of Theorem 8.6.1, this forces $\alpha_i = t_C$ and $\beta_j = -t_C$ on each component C of the design graph. The table is nonzero iff the t_C are not all equal, which is possible iff there are at least two components.